4 **(a)** Fig. 4.1 shows two metal containers, A and B.

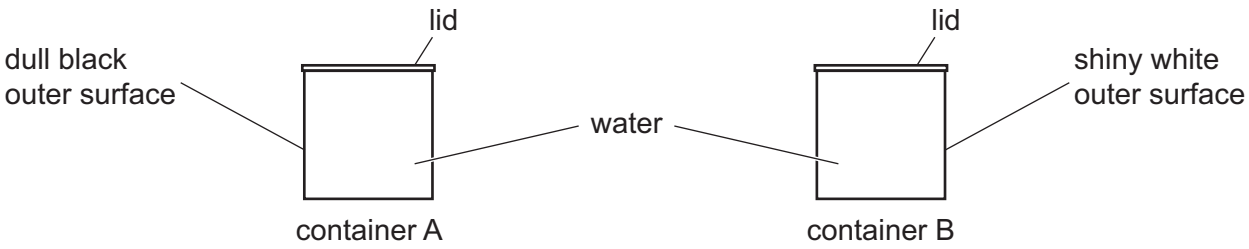

Fig. 4.1

Container A has a dull black outer surface and container B has a shiny white outer surface. A and B both contain very hot water at time = 0. The temperature of the water in each container is recorded every minute.

(i) State which container has the larger temperature change over the first 5 minutes. Explain your answer.

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..... [2]

(ii) After 30 minutes, the temperature of the water in container A remains constant.

State, in terms of energy transfers, why the temperature remains constant.

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..... [1]

(iii) The experiment is repeated without the lids on the containers.

State the name of the additional method of energy transfer that occurs.

..... [1]

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(b) Fig. 4.2 shows a metal pan with a wooden handle.

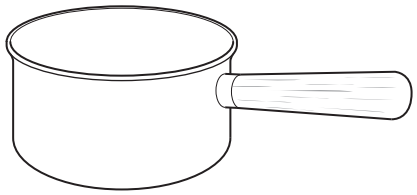

Fig. 4.2

Explain why there is a much greater rate of transfer of energy through the metal than through the wood.

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